QUANTUM TECHNOLOGIES AS STRATEGIC INFRASTRUCTURE: EVIDENCE, ADOPTION PATHWAYS, AND A NATIONAL RESEARCH AGENDA (2025-2035)

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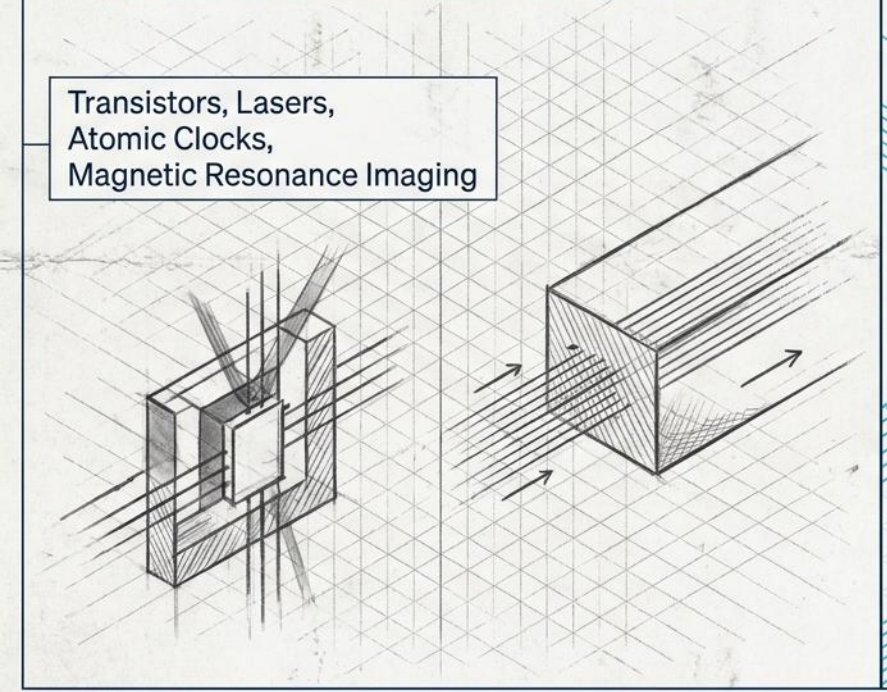
Quantum technology is transitioning from microscopic physics to macro-scale engineered infrastructure.

The First Quantum Revolution

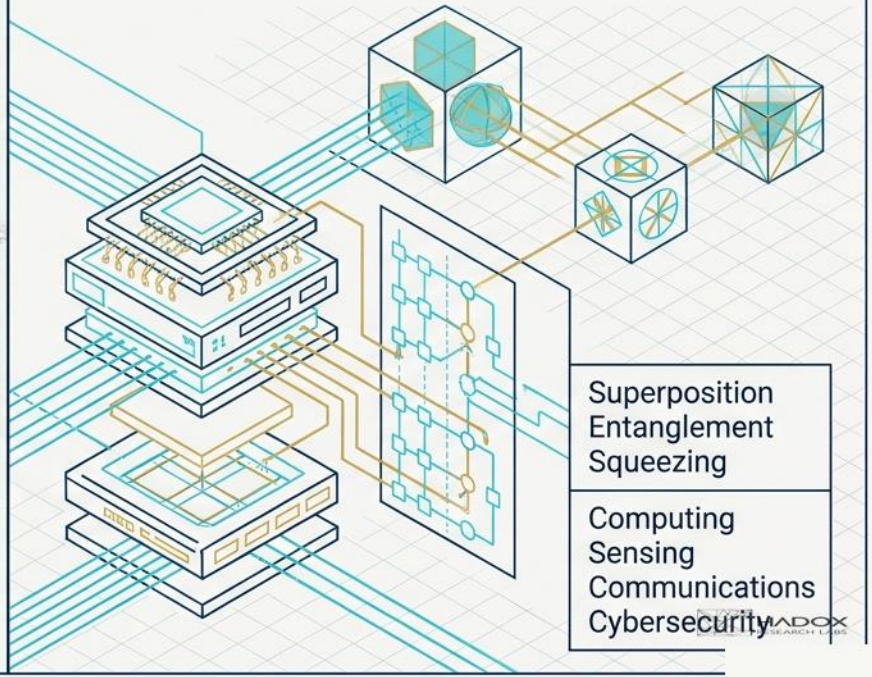
Bulk Properties & Statistical Phenomena

Transistors, Lasers,
Atomic Clocks,
Magnetic Resonance Imaging



The Second Quantum Revolution

Direct Engineering of Quantum States



Superposition
Entanglement
Squeezing

Computing
Sensing
Communications
Cybersecurity

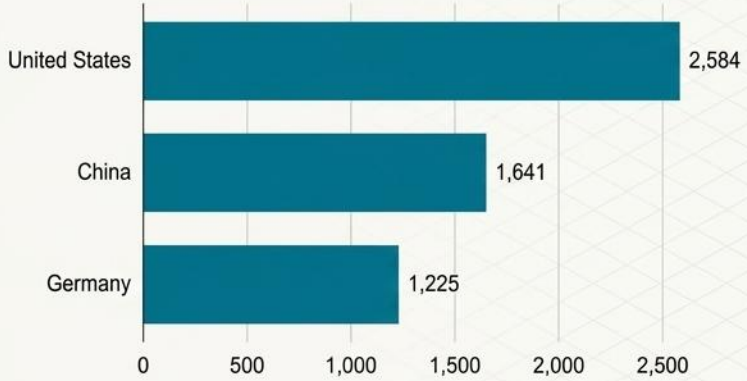
The empirical record reveals massive acceleration alongside structural and geographic unevenness.

$55.7B
Public Commitments worldwide (Oct 2025)

$11.235B
PPP in 12,209 public R&D awards (2015-2023)

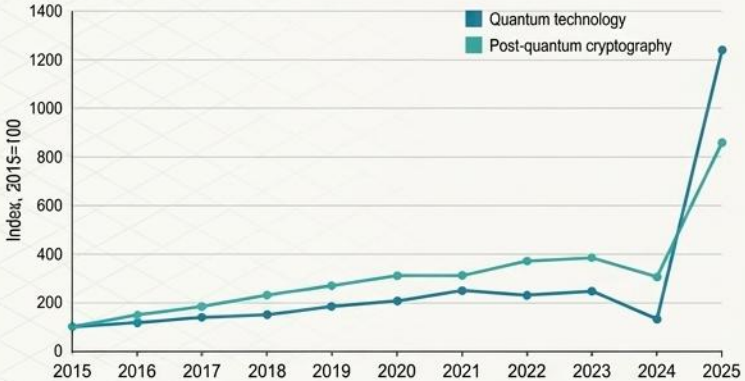
31,700
Quantum patent families (2005-2024)

Geographic Concentration of Scientific Capacity



Source: OpenAlex Works API extraction. Counts are bibliometric adoption proxies, not direct deployment measures.

Exponential Expansion of Literature



Source: OpenAlex Works API extraction. The 2025 quantum-technology spike is interpreted as an indexing/query signal requiring cautious interpretation.

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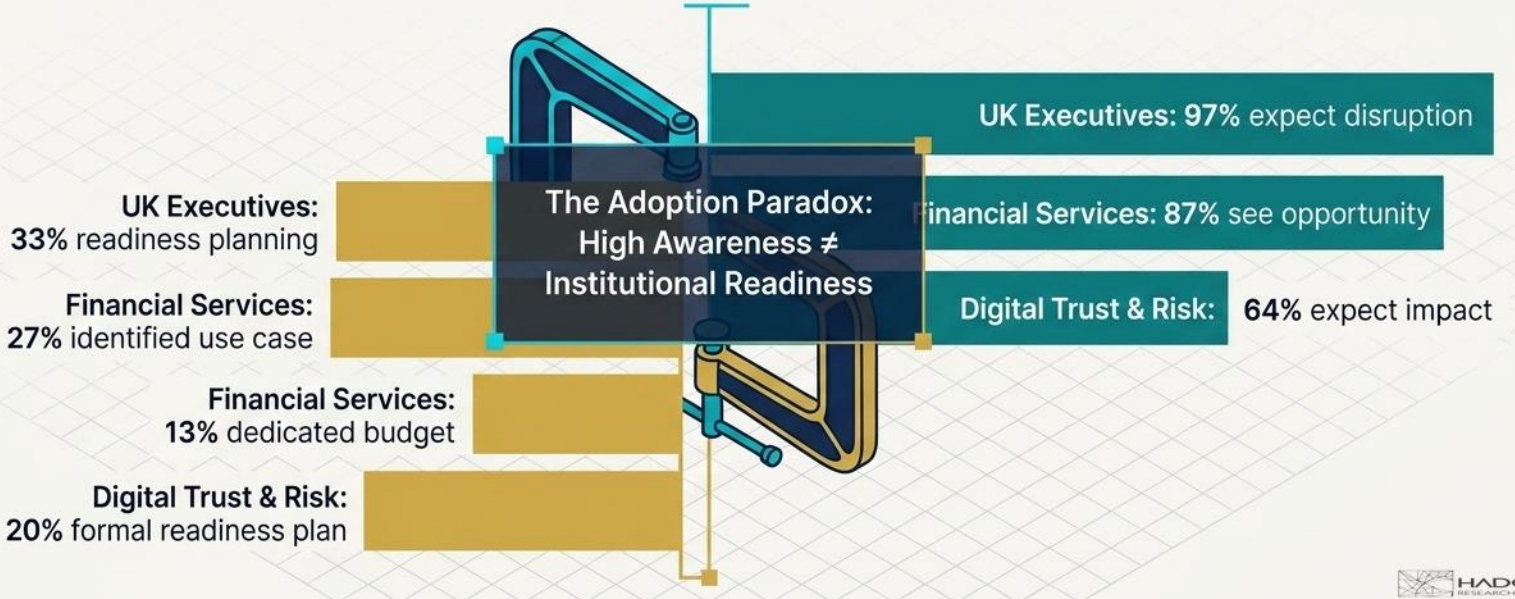
Editorial Blueprint

An extreme awareness-readiness gap currently paralyzes enterprise adoption.



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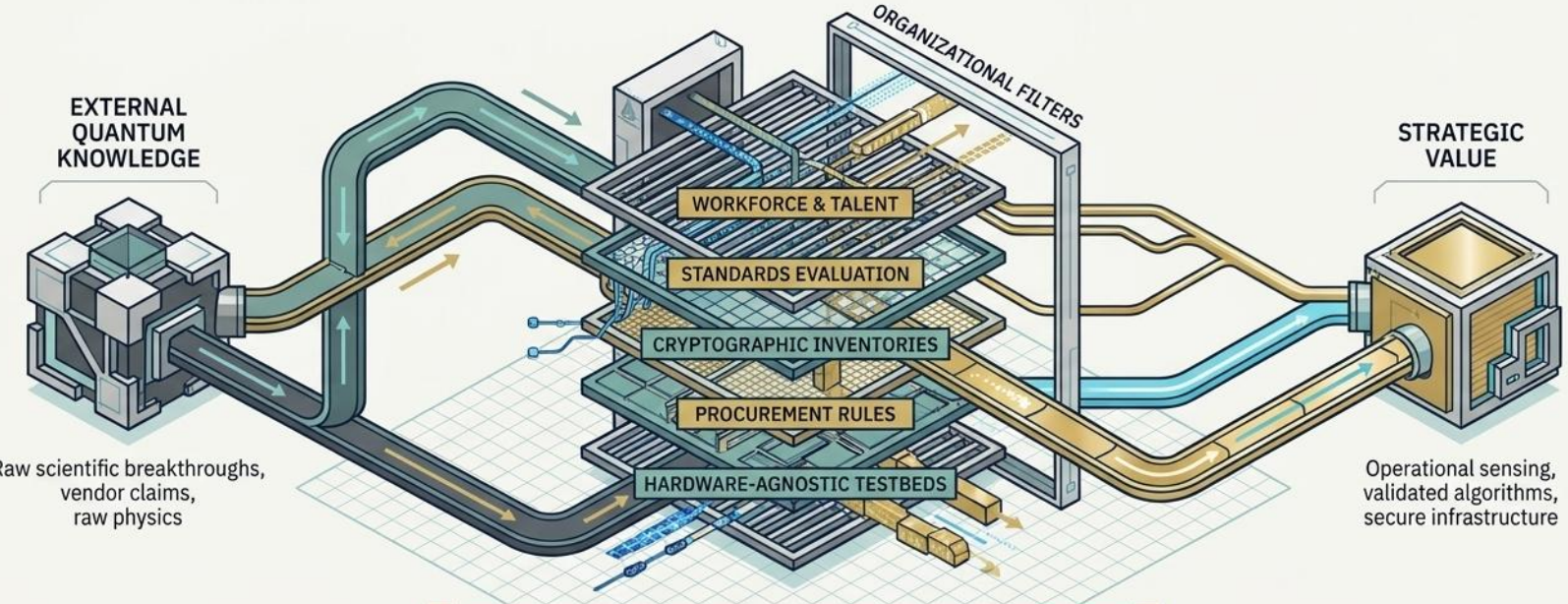
Quantum adoption is an absorptive capacity problem, not simply a hardware problem.

Absorptive Capacity (Cohen & Levinthal, 1990): The ability to recognize, assimilate, and exploit external knowledge.



RAW HARDWARE CAPABILITY CANNOT BYPASS ORGANIZATIONAL FILTERS.

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The quantum ecosystem forms a layered capability stack with distinct adoption timelines

Low Infrastructure Burden | High Infrastructure Burden

High Maturity

Post-Quantum Cryptography (PQC)
Immediate Security Mandate

Quantum Sensing
Near-Term Public & Industrial Value

Emerging Maturity

Quantum Computing
High Optionality, Low Near-Term Certainty

Quantum Communication
Specialized Infrastructure

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(Data Shelf Life + Migration Time) > Time to Quantum Computer = System Vulnerable Today

Harvest Now, Decrypt Later
Data collection happens in the present.

NIST 2024 Standards Finalized
ML-KEM, ML-DSA, SLH-DSA

Time to Quantum Computer

Post-Quantum Cryptography (PQC) is the first undeniable digital-infrastructure transition.
PQC requires multi-year cryptographic inventories, crypto-agility, and legacy system replacement.

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Strategic evaluation requires shifting metrics from physical qubits to logical-qubit economics

Metric Shift:
Physical Qubits → Logical-Qubit Economics

	Strength	Bottleneck	Strategic View
Superconducting	Fast gates, industrial ecosystem	Cryogenic wiring, thermal load	Scale depends on error correction
Trapped Ions	High fidelity, long coherence	Gate speed, shuttling	Engineering scale/throughput remain central
Neutral Atoms	Large arrays, flexible geometry	Atom loss, gate fidelity	Strong option for simulation
Photonics	Network compatibility, room-temp	Photon loss, deterministic gates	Crucial for networking
Annealing	Available systems	Limited universality	Must be benchmarked against classical heuristics

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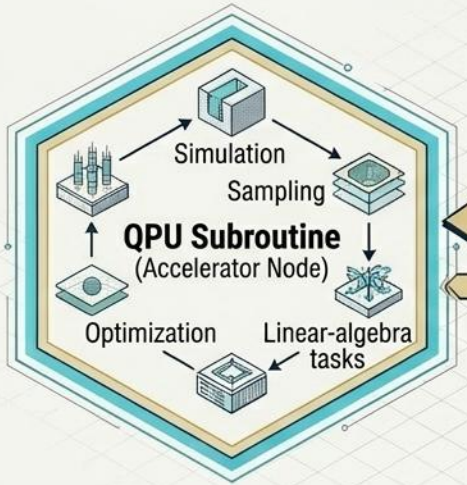
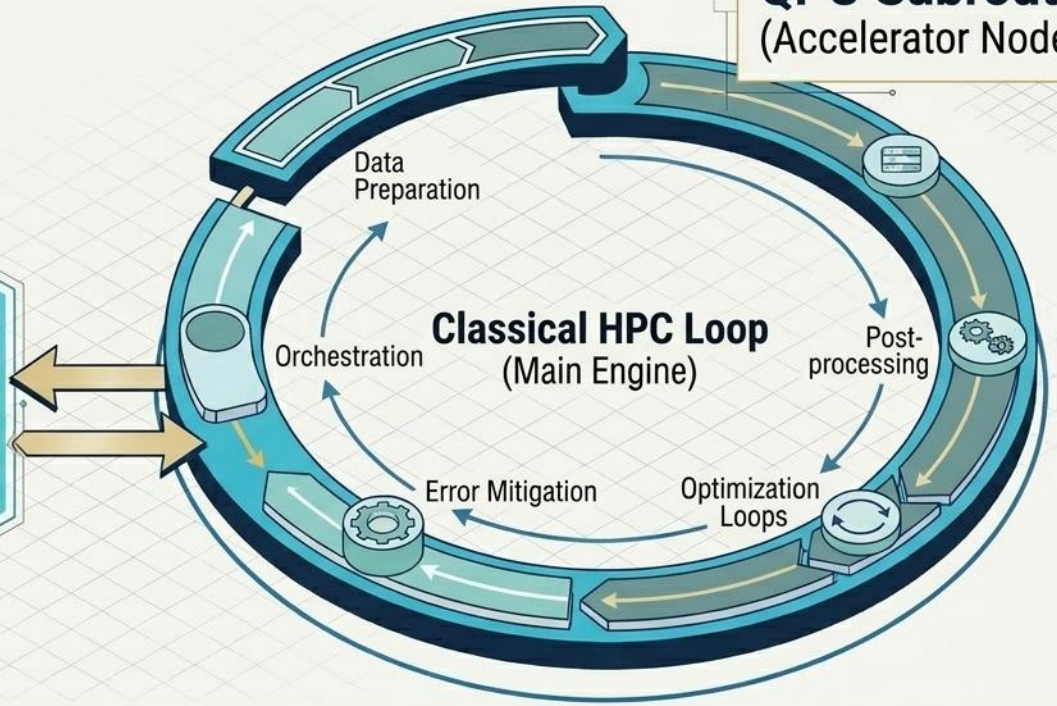
Architectural Workflow





The strategic objective is an HPC-QC continuum where supercomputing remains the operational substrate, with QPUs invoked under realistic latency, noise, and verification constraints.

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Scientific relevance does not automatically guarantee enterprise or industrial adoption.

Chemistry & Materials

Strongest Long-Run Case

Simulates quantum systems naturally, but requires hybrid algorithms and integration with lab workflows.

Optimization

High Hype, Difficult Baselines

Must beat exceptionally strong classical mixed-integer programming and GPUs. Currently exploratory.

Finance

Demanding Viability

Must improve speed, regulatory auditability, and model quality compared to classical Monte Carlo.

Machine Learning & AI

Research Stage

Suffers from barren plateaus. AI is more likely to improve quantum systems than vice versa right now.

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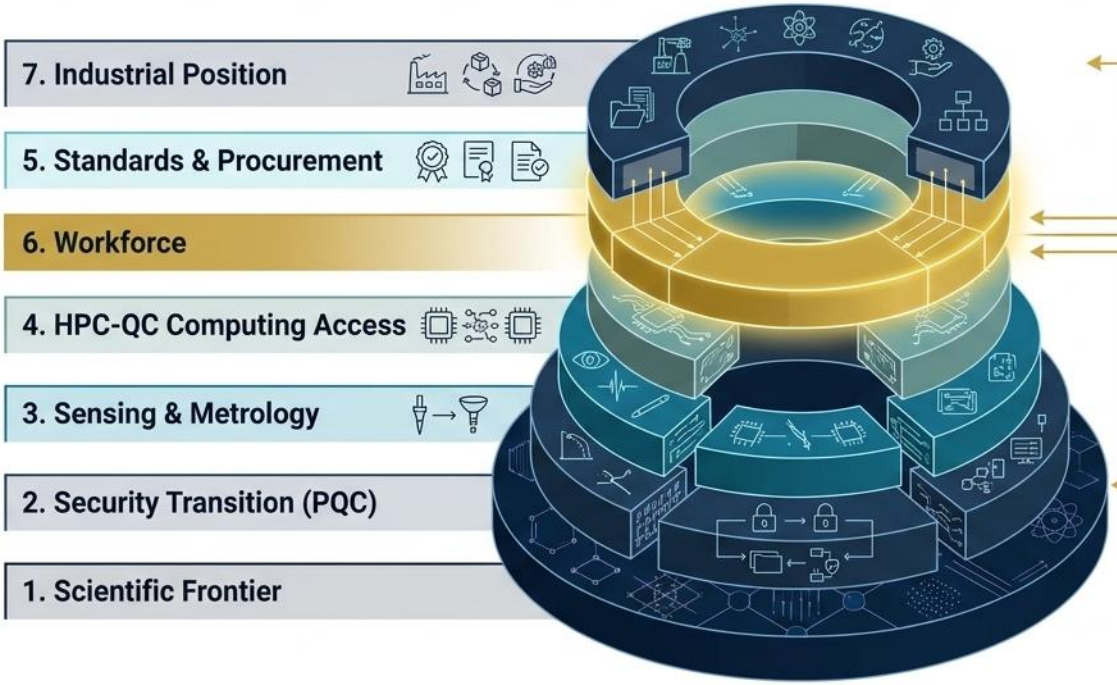
A seven-pillar capability model separates operational readiness from rhetorical innovation policy



The Missing Middle

A nation cannot rely only on physics PhDs; it needs technicians, control system engineers, cybersecurity auditors, and procurement officers to translate science into infrastructure.

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National Quantum Roadmap and Strategic Capability Sequencing, 2025-2035

Capability accumulation proceeds through staged preparation, operational testbeds, and selective scaling. Roadmap phases represent capability milestones, not hardware-breakthrough predictions.

2025-2027 Preparation & Security

- PQC inventories, QCaaS access, workforce curricula, sensing feasibility

2027-2030 Testbeds & Pilots

- Sensing pilots, PQC staged migration, benchmark labs, justified QKD

2030-2035 Selective Scaling

- Operational sensing, quantum networks, logical-qubit readiness, frontier research

	2025-2027 Preparation	2027-2030 Testbeds	2030-2035 Scaling
PQC	3	3	3
Sensing	2	3	3
QCaaS	2	3	2
Communication	1	2	3
Standards	2	3	3
Workforce	3	3	3
Frontier R&D	2	2	3

not central | watch | build | priority

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Six strategic errors consistently threaten enterprise decision-makers and policy scholars.

1. Waiting

Delaying action until fault-tolerant machines arrive ignores urgent PQC and sensing mandates.

2. Hardware Nationalism

Buying a domestic computer without a trained workforce creates a symbolic asset, not a capability.

3. Confusing QKD with PQC

Treating Quantum Key Distribution as a broad substitute for PQC misreads the infrastructure problem.

4. Treating QCaaS as Neutral

Cloud access democratizes experimentation but concentrates platform power and data governance.

5. Baselineless Advantage Claims

Accepting speedup claims without specifying the classical baseline, error model, and economic relevance.

6. Underfunding the Missing Middle

Ignoring the technicians and engineers needed to operate the systems.

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Quantum readiness is a cultivated institutional capability, not a hardware purchase.

Quantum strategy succeeds when it turns fragile quantum effects into reliable public capability.

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Evidence base and source repository

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Repository:
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